Development Setting and Methods

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Research-based Development

- Laurea's theses are research-based development projects
- Work-life orientation
- Various approaches (e.g., case study, service design)
- Utilization of different methods
  (e.g., interviews, observation, surveys, document analysis)
- A development process that results in an outcome
  beneficial to working life

The Nature of Research-based Development Work (Moilanen et al. 2022)

Scientific research	Research-based development	Development based on everyday thinking
▪ Aims to create new theory and test existing theories ▪ Follows the tradition of scientific research: Reflects on philosophical questions of science and seeks answers to the research problem using generally accepted methods. The theoretical framework indicates which scientific discussion is being engaged with. The results are published in a research publication intended for the scientific community. ▪ The researcher can be "detached" from the research subject, and interaction may be minimal during the research process.	▪ Aims to solve problems arising from practice or reform practices, often also generating new knowledge about workplace practices. ▪ Information is systematically collected and critically evaluated from both practice and theory to support development. ▪ A wide range of different methods is used. ▪ Active interaction with various stakeholders is emphasized. ▪ Writing and presenting at different stages and for different audiences drive the development work forward.	▪ Aims to solve problems arising from practice or to reform practices. ▪ Information is collected from practice and theory randomly. ▪ Decisions are largely based on personal ideas with little justification. ▪ Critical evaluation is minimal. ▪ Interaction is limited. ▪ If reporting is done, the report is written only after everything is completed.

Development Setting

- The development setting is a section of Laurea's master's thesis.
- It is a separate component related to the development of the thesis, distinct from the theoretical framework.
- This section describes the development process of the thesis before presenting the results.
- Thesis in Master's studies

Develoment Setting

- Purpose, objectives, development tasks, and development questions
- Methodological solutions
- Description of the development method and how it is used in the thesis
- Data collection: what kind of data is collected and how it is collected
- Data management plan (where the data is stored, who has access to it, how it is disposed of)
- Description of data analysis: how the data is analyzed

Different Approaches to Development

- Case study
- Constructive research
- Action research
- Service design
- Foresight
- Innovation creation
- Many of these approaches are closely related to each other.

Research-based development work process (Moilanen et al 2022)

1. Identification of the development target and setting preliminary targets
2. Searching data related to the development target both from practice and by getting familiar with the existing theory
3. Defining the scope
4. Writing the literature review, planning the approach and methods
5. Implementing the development, sharing results in different forms
6. Evaluating the development work

Methods in Research-based Development

- Various methods can be utilized in different stages of development.
- Mixed method
- For example, a survey is conducted with consumers at the beginning, followed by interviews with the organization's employees.
- For example, relevant documents are studied using document analysis, and based on them, something is developed. Later, an interview/survey/workshop is conducted, where observation takes place.

Qualitative Research:

- Mapping people's understanding of the desired question or problem
- Questions/methods may change as the research progresses
- A research approach that emphasizes the importance of individuality and diversity

Quantitative Research:

- Testing an objective theory by examining relationships between variables using statistical methods
- Variable are predefined and numerically measurable
- A research approach that emphasizes generalizability and the exclusion of alternative explanations

Qualitative:

- Aims to understand the quality, characteristics, and meanings of the subject holistically
- Instead of predefined questions by the researcher, participants are relatively free to discuss topics related to the subject

Quantitative:

- Based on describing and interpreting the subject using statistics and numerical data

When is Quantitative Research Used?

- What kind of information is desired to be produced?
- Where can the necessary data be obtained from?
- Large datasets
- Numerical data
- Survey research
- Experimental research: experimental and control groups

When to Use Quantitative Research?

- One possible starting point could be: What would be the best way to find an answer to the problem/question?
- For example, if the topic is improving the customer experience in an employee restaurant:
  - The aim is to assess the current state of customer experience and determine how satisfaction with products and services can be improved.

Quantitative Research

- What is the current customer experience? A survey is conducted at the starting point.
- Changes are made, meaning something is developed. After this, a new survey is conducted.
- Using statistical methods, it can be determined whether there is a difference in customer experience before and after the changes.

Quantitative Research

- Large sample size
- Statistical analysis based on numerical data
- Surveys
- Measurements

Sample

- The sample is assumed to represent the population.
- Conclusions about the entire population are drawn based on the sample results.
- In a census study, the entire population is examined

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Appropriate Sample Size

- Depends on how large a population the results are intended to be generalized to.
- Also depends on the representativeness of the sample.
- For comparisons, there must be a sufficient amount of data.
- The more detailed the information needed, the larger the sample should be.

- Typically, at least 50 respondents in a survey study.

Causal Relationship

- The connection between an explanatory variable and a dependent variable (= causality).
- How is event A related to measurable outcome B? (A = an advertisement is made → B = sales increase)
- Does something else influence B?

Causal Relationship

- A sufficiently large dataset: the bigger, the better.
- A normally distributed dataset for parametric tests.
- How many different tests are needed, or are mean distributions sufficient?
- What kind of questionnaire is suitable for quantitative research?

Survey

- People's behavior, decisions, and attitudes are examined using a set of questions.
- A single survey, cross-sectional study: descriptive.
- Multiple surveys, longitudinal study: better possibility for assessing cause-and-effect relationships.
- A sample is selected from the target population, and the observations made are intended to be generalized to the entire population.

Survey

Before starting to create the survey questionnaire, the following should be ready:

- Research problem(s)
- Sufficient theoretical knowledge about the subject
- Research variables

Survey Questions/Statements

- Only ask questions that are necessary for the research.
- The form and questions should be kept as simple as possible.
- Everyone should answer the questions consistently.
- How does the information obtained from the question help answer the research questions?
- Can you truly obtain the information you want with the question?
- Could the same information be obtained more effectively or reliably with a different type of question?

One Example of Likert-Scale Statements



Another Example of Survey Statements

	Never	Rarely	Some of the time	Most of the time	Always	Does not apply to me
a. I wash my hands with soap and warm running water before preparing food.	☐	☐	☐	☐	☐	☐
b. After playing with a pet and before getting a snack, I wash my hands with soap and warm running water.	☐	☐	☐	☐	☐	☐
c. After cutting raw meat, chicken, or seafood, I wash all items that came in contact with the raw food (e.g., cutting board, knife, counter top) with hot, soapy water before I continue cooking.	☐	☐	☐	☐	☐	☐
d. I thoroughly rinse fresh vegetables under running water before eating them.	☐	☐	☐	☐	☐	☐
e. I wash the plate used to hold raw meat, poultry, or seafood with hot, soapy water before returning cooked food to the plate OR I use a clean plate.	☐	☐	☐	☐	☐	☐
f. I wash my hands with soap and warm running water after working with raw meat, chicken, or seafood and before I continue cooking.	☐	☐	☐	☐	☐	☐
g. When I cook fish, I check that the flesh flakes easily with a fork before serving.	☐	☐	☐	☐	☐	☐
h. I store my eggs at room temperature.	☐	☐	☐	☐	☐	☐

Distributions, Frequencies, and Measures of Central Tendency

- A common way to present the results of a survey. If there is sufficient data, averages can be calculated.
- Additionally, mode and median can also be calculated.
- The results can be presented in the form of bar charts, pie charts, or tables.

Qualitative Research

Qualitative Research

- Qualitative research typically involves interviews, sometimes also observation, document analysis, and electronic surveys.
- Qualitative data is usually verbal, but it can also be processed numerically through quantification.

The Process of Qualitative Research

1. Planning
2. Familiarization with the topic
3. Data collection
4. Data analysis
5. Drawing conclusions, reporting results and the process

Different Types of Qualitative Data

- Interviews
- Texts: books, biographies, articles, reports, discussion forums, and social media
- Videos of various situations and events
- Observation data

Interview

- Thematic interviews are the most common type of interview in thesis projects.
- This is a so-called semi-structured interview, where the interview is guided by predetermined themes, often emerging from previous research. Other forms of interviews include, for example, in-depth interviews (open interviews) and focus group interviews (group interviews conducted for a specific target group).
- Note that the themes of the interview are not the same as the themes used in content analysis. Usually, content analysis is data-driven.

What do I do with the data?

- Don't panic about the large amount of data! There should be a lot of data in order to gather enough information for drawing conclusions.
- Read the data several times and think about what it reveals and what can be learned from it. Also, consider how it relates to the literature you have studied during your studies.
- Start shaping your story: What do I want to know, how do the parts of the data relate to each other, and how is the data linked to the theoretical background?

The data can be analyzed using, for example, content analysis.

- **Data-driven:** This is the most common form of content analysis in thesis projects. The data guides the analysis. The analysis units are selected from the data in accordance with the purpose of the research; they are not predefined.
- **Theory-guided:** Theory assists in the analysis, but the analysis is not directly based on theory. The analysis units are selected from the data, but previous knowledge guides the analysis. Theory helps in thinking, but theory is not tested in the analysis. The terminology comes from the theory.
- **Theory-driven** The data is analyzed based on a theoretical framework. The theory or conceptual system serves as the reference framework that guides the analysis.

Another Analysis Types of Qualitative Research

- Typification
- Thematization
- Quantification

Content Analysis

- The goal is to reduce the data (trim it) and thus find the core meanings of the text.
- Three Stages of Content Analysis:

1.Data Reduction: Irrelevant data is removed from the dataset for the research.

2.Clustering the Data: The original expressions in the data are reviewed, and concepts describing similarities or differences are sought. These concepts are then grouped into categories, from which subcategories are created.

3.Data Abstraction: Irrelevant information for the research is separated from the data, and theoretical concepts are formed based on the selected data.

In the conceptualization phase, the process moves from the expressions used in the original data to theoretical concepts and conclusions.

- In development work, this phase can mean identifying concrete problem areas.

Literature

Moilanen, T., Ojasalo, K., and Ritalahti J. 2022. Methods for Development Work. New kinds of competencies in business operations. Books on Demand GmbH. Available in Laurea Library as book or e-book.

Questions?